Employee ID Card Request
Capstone Project Documentation

1. Introduction

- **Project Title:** Employee ID Card Request System using ServiceNow
- **Purpose:** To automate the submission, approval, and issuance process for employee ID cards using the ServiceNow platform.
- **Features:** ID card request creation via Service Catalog, automated approval workflows, task generation for admin teams, and notification delivery.

2. Architecture

The system follows a layered architecture to ensure scalability and maintainability.

- **Presentation Layer:** Uses the ServiceNow Service Portal to provide a user-friendly interface for employees to submit requests.
- **Application Logic Layer:** Employs Business Rules to automate workflow routing upon submission and Script Includes to handle reusable logic, such as data validation and notification triggers.
- **Data Layer:** Utilizes ServiceNow tables (e.g., `sc_req_item` or a custom table) to store structured data like employee information, department details, and request status.

3. Setup Instructions

- **Create Service Catalog Item:** Define catalog variables such as Employee Name, Department, Request Reason, and an attachment field for the ID photo.

- **Configure Backend Automation:** Set up Business Rules to trigger manager approval workflows and generate tasks for the security or HR team once approved.
- **Configure Notifications:** Set up automated email notifications to inform employees of their request status (e.g., "Approved" or "Pending").
- **Reporting & Dashboard:** Build dashboards using ServiceNow Reporting tools to track the number of requests, processing times, and pending tasks.

4. Testing

The system should be validated using a structured manual testing approach.

- **Test Scope:** Testing should cover the Service Catalog request form, approval routing, task generation, and notification delivery.
- **Sample Test Scenarios:**
    - **Request Submission:** Verified that valid requests create records in ServiceNow.
    - **Input Validation:** Ensured all mandatory fields are filled before form submission.
    - **Workflow Execution:** Confirmed that requests are correctly routed to the manager for approval.
    - **Notification Delivery:** Confirmed that email alerts are sent to the requester upon status changes.

5. Known Limitations & Future Enhancements

- **Known Issues:** The system relies on internal ServiceNow workflows and lacks integration with physical printing hardware or advanced biometric verification.

- **Future Enhancements:** Future iterations could include integration with physical ID card printing machines, real-time status tracking via a mobile app, or automated facial recognition for badge generation.

6. Running the Application

The system operates through an automated ServiceNow workflow.

- **Step 1: Access the Service Portal:** Employees open the portal via the instance URL and navigate to the "Employee ID Card Request" catalog item.
- **Step 2: Fill Request Form:** The user enters required details like Department and Reason, and attaches their photo. Client-side validations (Client Scripts/UI Policies) ensure all mandatory fields are completed before submission.
- **Step 3: Submit Request:** Once submitted, a service request is generated, and a record is created in the RITM or custom ID card table.
- **Step 4: Backend Processing:** Business Rules automatically trigger after record creation to handle logic such as routing for manager approval and finalizing metadata.
- **Step 5: Notification Delivery:** Once processing is complete (e.g., manager approval), an automated email notification is triggered to update the employee.
- **Step 6: Monitoring and Verification:** Admins track requests through ServiceNow record views or dashboards showing status and analytics.

7. Automation Logic

The system relies on ServiceNow’s internal automation framework.

- **Internal Automation:** Operations such as triggering Business Rules on record creation, managing approval workflows, and sending notifications are handled natively within the platform to ensure the process remains fully automated.
- **Automation Trigger Point:** The workflow process is initiated by an "after insert" Business Rule, ensuring that actions only occur after a valid request is successfully recorded.

8. Authentication

The project leverages ServiceNow’s built-in Role-Based Access Control (RBAC).

- **Authentication Mechanism:** Users must log in with valid credentials, with session management and identity verification handled by the platform’s security layer.
- **Role-Based Access Control:** Access is restricted by roles; for example, the "User" role allows employees to view their own requests, while the "Admin" role provides full access to configuration and settings.
- **Data Security:** Security rules (ACLs) apply table-level and record-level security, ensuring users can only view their own requests.

9. User Interface

The system interface is designed for simplicity and efficiency.

- **Request Form:** A clean, responsive Service Catalog form is used for data entry, featuring dynamic validations and error prompts to prevent incorrect submissions.
- **Confirmation & Dashboard:** Users are redirected to a confirmation page post-submission, and administrators use reporting tools to view total requests, status tracking, and request volume trends.

10. Testing

The system was validated using a structured manual testing approach.

- **Test Scope:** Testing was performed across core modules, including the Service Catalog request form, Business Rules execution, approval workflows, and notification delivery.
- **Test Cases:** Scenarios such as valid request submission, input validation (e.g., mandatory field checks), and notification delivery were tested with "Pass" results.

11. Known Issues

- **Limited Hardware Integration:** The system does not integrate with physical ID card printing machines.
- **Connectivity Dependency:** Users require active connectivity to ServiceNow for request submission and status tracking.
- **Basic Fraud Protection:** The system does not use advanced encryption or blockchain-level validation for digital ID cards.
- **Scalability:** The system has not been tested under high-load conditions (large-scale concurrent requests).

12. Future Enhancements

- **Printing Integration:** Integration with physical card printing systems for automated badge production.
- **Mobile App:** Enabling status tracking and request submission via a mobile platform.